

Oracle Architecture Notes

Instance Architecture

- Instance equals SGA memory plus background processes.
- Shared Pool handles parse and dictionary metadata; Buffer Cache handles data blocks; Redo Buffer stores change vectors.
- PGA is private process memory for sort hash and runtime state.

Process Responsibilities

- DBWn writes dirty buffers to datafiles.
- LGWR flushes redo on commit boundaries.
- CKPT updates control-file and datafile checkpoint metadata.
- SMON and PMON handle recovery and process cleanup.
- ARCn archives filled online redo logs.

Storage Structures

- Datafiles store persistent table and index blocks.
- Control files store structural metadata and checkpoint history.
- Online redo logs and archived redo logs provide crash and media recovery chain.
- Undo and temporary segments support rollback read-consistency and sort/hash workloads.

Diagnostic Commands

- `SELECT instance_name, status, database_status FROM v$instance;`
- `SELECT name, description FROM v$bgprocess WHERE paddr <> '00' ORDER BY name;`
- `SELECT file#, name, status FROM v$datafile ORDER BY file#;`
- `SELECT group#, status, archived FROM v$log ORDER BY group#;`